

CN ASSIGNMENT 01

SECTION-A

Q1: Total DNS Resolution time = Root DNS time + TLD time + Authoritative Server time

$$T = 60 + 50 + 40 = 150 \text{ ms}$$

$$\text{Time saved via cache} = T - \text{Auth time} = 150 - 40 = 110 \text{ ms}$$

Q2: At 10:00³³ AM, Yes ... total time saved = $150 - 2 = 148 \text{ ms}$

At 10:07 AM, No ... as 5 minute TTL expired.

Q3: File size = 12 MB = $12 \times 8 = 96 \text{ megabits (Mb)}$

$$\text{Now link 1, } T_{d1} = \frac{96}{20} = 4.8 \text{ s, Link 2, } T_{d2} = \frac{96}{100} = 0.96 \text{ s}$$

$$\text{Total Transmission delay: } T_d = T_{d1} + T_{d2} = 4.8 + 0.96 = 5.76 \text{ s}$$

Q4: Total end to end delay = $\sum (T_d + T_p)$

$$\text{Link 1, } T_d = \frac{8000}{40 \times 10^6} = 0.2 \text{ ms} \text{ \& } T_p = 20 \text{ ms} \dots T_1 = 20.2 \text{ ms}$$

$$\text{Link 2, } T_d = \frac{8000}{80 \times 10^6} = 0.1 \text{ ms} \text{ \& } T_p = 30 \text{ ms} \dots T_2 = 30.1 \text{ ms}$$

$$\text{Link 3, } T_d = \frac{8000}{500 \times 10^6} = 0.016 \text{ ms} \text{ \& } T_p = 5 \text{ ms} \dots T_3 = 5.016 \text{ ms}$$

$$T = T_1 + T_2 + T_3 = 20.2 + 30.1 + 5.016 = 55.316 \text{ ms}$$

$$\text{Q5: a) Total traffic without cache} = 5000 \times (400 \times 10^3) = 2 \times 10^9 \frac{\text{bits}}{\text{hr}} = 2 \text{ Gb/hr}$$

$$\text{b) Total traffic with cache} = 5000 \times (400 \times 10^3) \times (0.25) = 0.5 \text{ Gb/hr}$$

$$\text{c) Total traffic saved} = 2 - 0.5 = 1.5 \text{ Gb/hr}$$